Project Title

(Logistics Optimization Using Minimum-Cost Maximum-Flow)

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

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ABSTRACT

Logistics Optimization Using Minimum-Cost Maximum-Flow:

This project presents a logistics optimization system designed to minimize transportation costs while satisfying the supply available at warehouses and the demand requirements of destinations. It models warehouses, destinations, and transportation routes as a flow network, where capacities represent shipment limits and costs represent transportation expenses. The Minimum-Cost Maximum-Flow (MCMF) algorithm determines the optimal shipment quantities along the most cost-effective routes while ensuring all supply and demand constraints are satisfied. The system displays the selected transportation routes, quantities shipped, individual route costs, and total transportation cost through a detailed cost report. It also provides a visual representation of the transportation network to make the optimized flow easy to understand. The project demonstrates the practical application of graph algorithms and cost-based network optimization in supply chain and logistics management.

Signature of the Guide